

EVALUATING THE IMPACT OF RISING EMERGING MARKET CORPORATE DEBT

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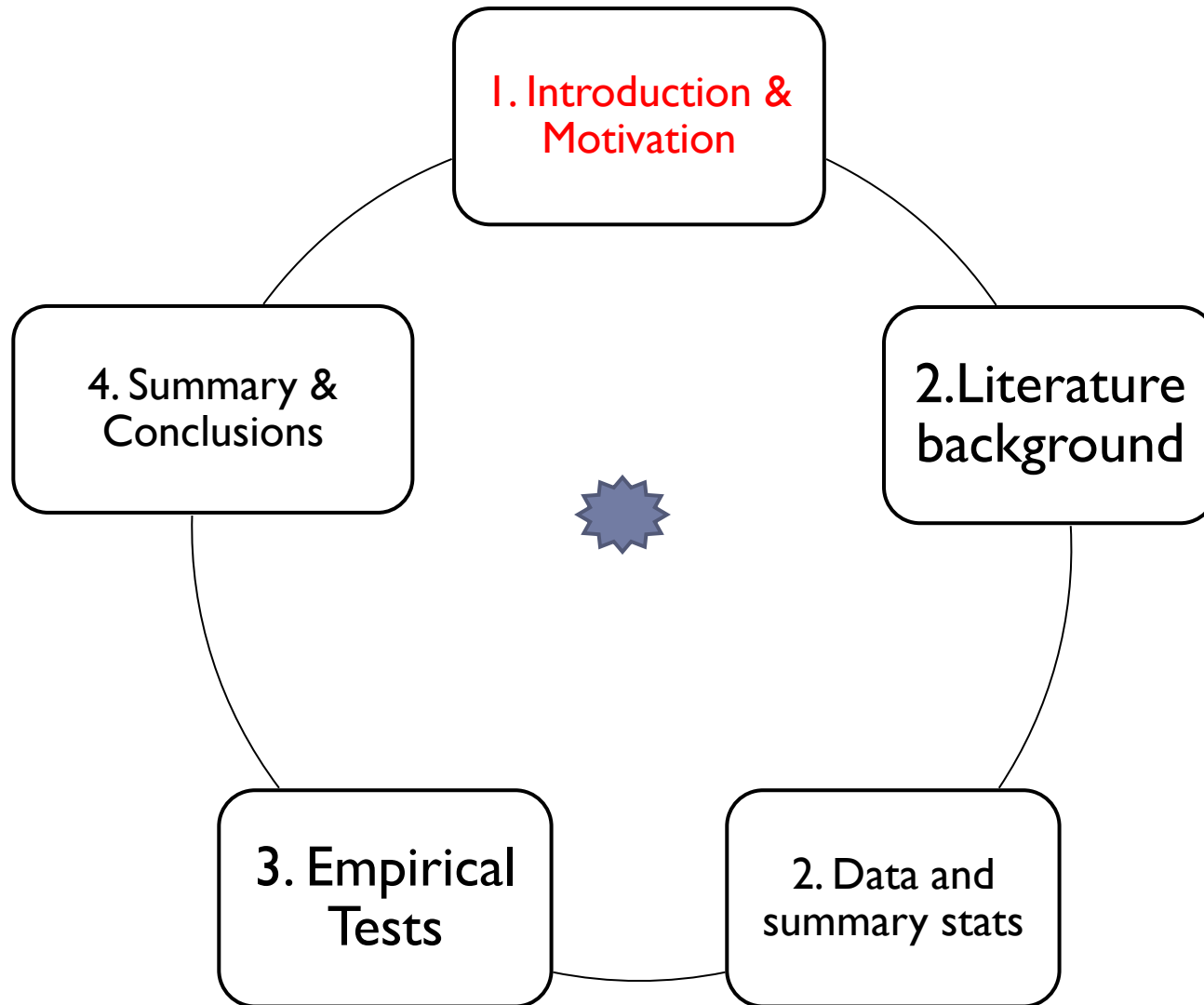
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8th Emerging Markets Finance Conference, Mumbai, 2017



Agenda

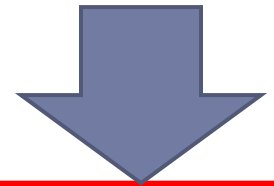


The story...

The
Economist

The world economy

The never-ending story



First America, then Europe. Now the debt crisis has reached emerging markets

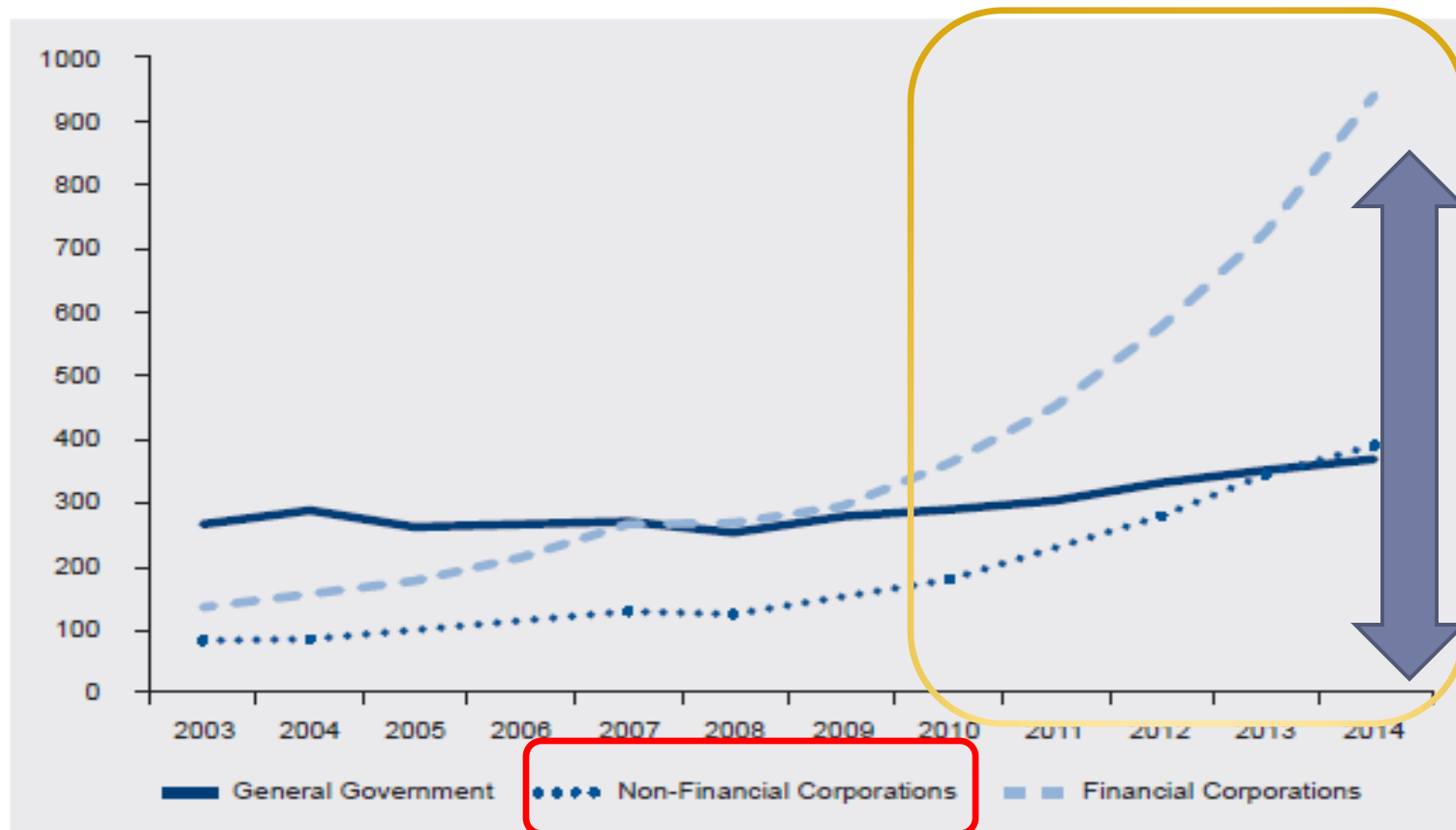
Nov 14th 2015 | From the print edition

Motivation:

Key aspects of emerging market debt

- I. Emerging market corporate loans and debt rose from
 - ▶ 73% of GDP at the end of 2007 to
 - ▶ 107% (or 127% if we include shadow banking debt) of GDP by the end of 2014.
- II. Offshore issuance of corporate bonds in foreign currency mainly by non-financial corporations has resulted in
 - ▶ currency mismatch on the consolidated balance sheets of emerging market firms (Shin, 2013)
 - ▶ increased borrower's interest rate, rollover and currency risks (Chui, Fender and Sushko , 2014)
- III. Having obtained funds abroad, the foreign affiliate of a non-financial corporation normally transfer funds to its home country via three channels (Avdjiev, Chui and Shin, 2014):
 - ▶ it could lend directly to its headquarters (within company flows),
 - ▶ extend credit to unrelated companies (between-company flows) or
 - ▶ make across-border deposit in a bank (corporate deposit flows).

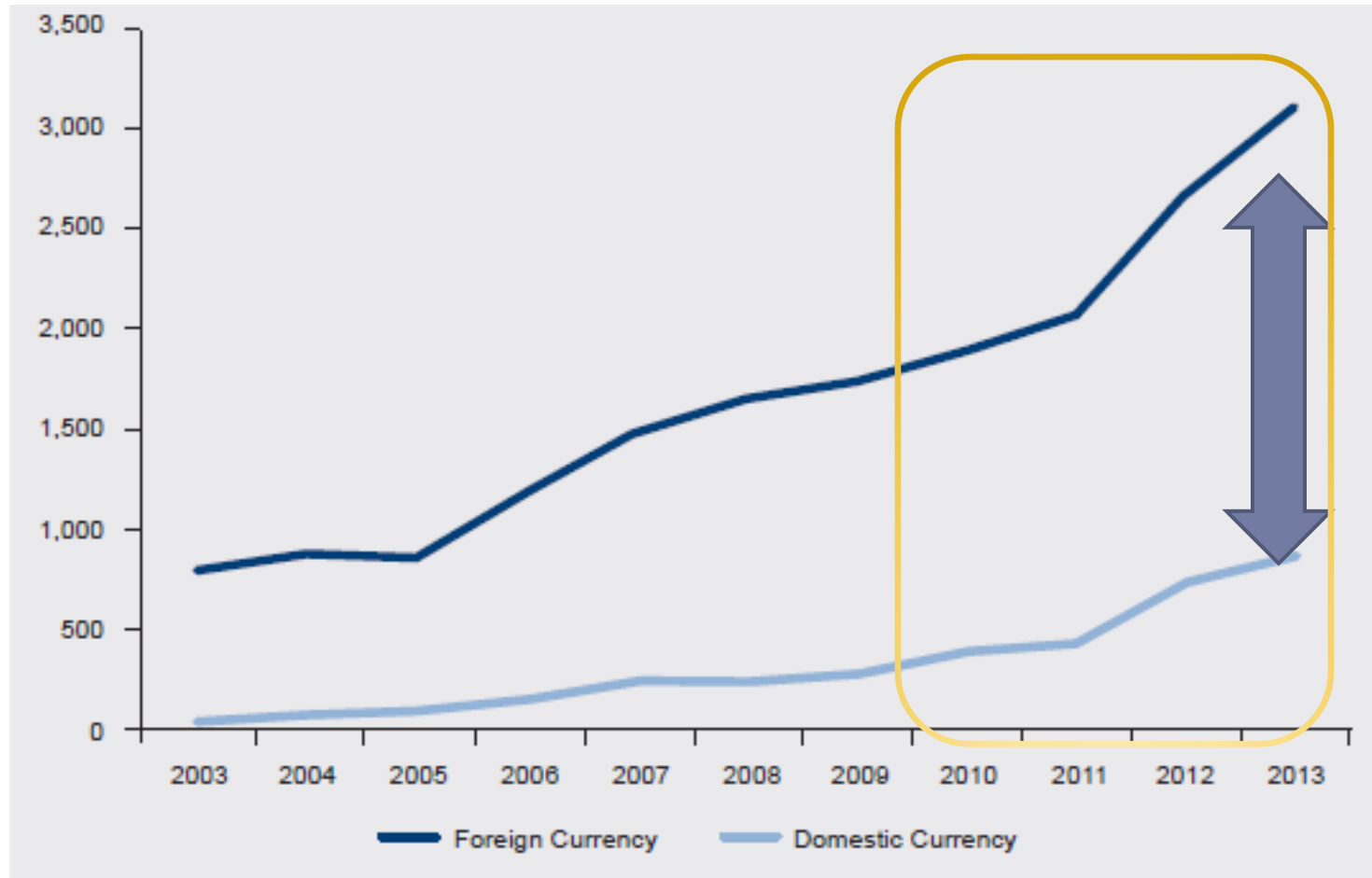
International bond issuance by Emerging Market Economies (€ Bi)



Note: Emerging market includes Argentina, Brazil, Chile, China, Colombia, India, Indonesia, Israel, Korea, Malaysia, Mexico, Philippines, South Africa, Thailand, and Turkey. Financial corporations include banks and other financial corporations.

Source: BIS.

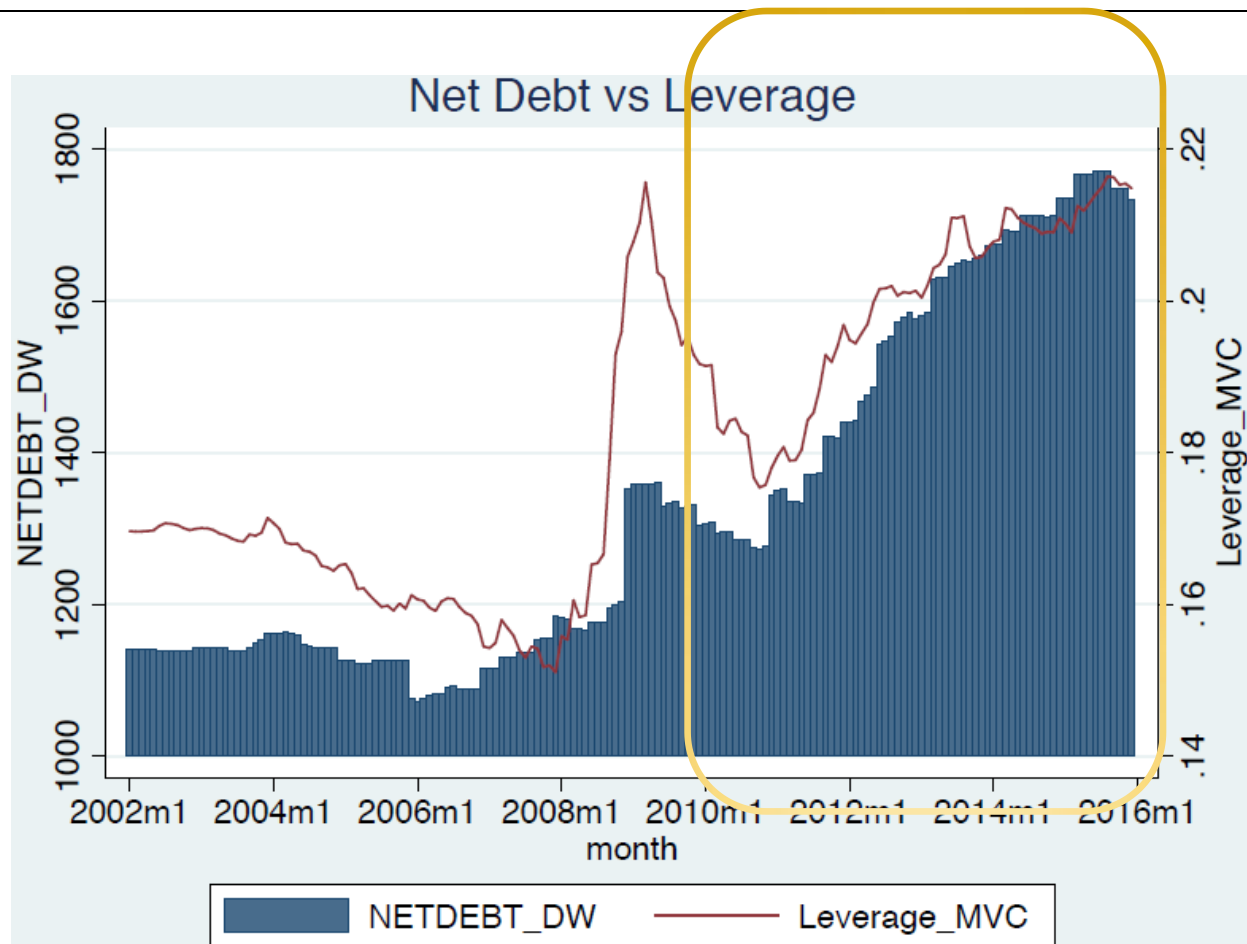
Currency Composition of Outstanding Emerging Market Bonds



Note: Billions of U.S. dollars on vertical axis.

Source: World Bank database

Post crisis growth in emerging market corporate leverage (Source DataStream)



Lot of Cross-sectional variation..

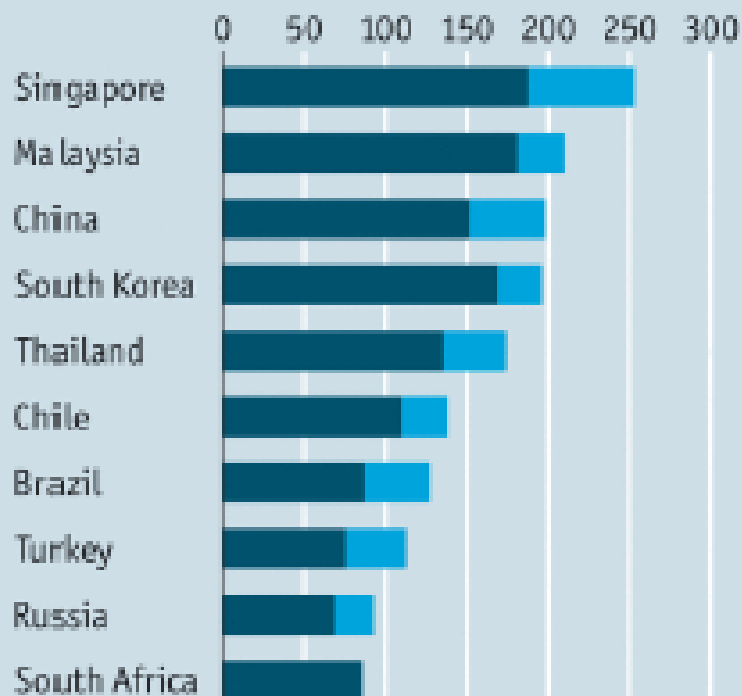
Big and bigger

2

Emerging-market private debt
As % of GDP

2007

2014



Pulled back in (The Economist)

India

Indonesia

Mexico

Argentina

Sources: J.P. Morgan; BIS; IMF

In this paper..

- ▶ We investigate the potential fallout from the post-crisis increase in corporate leverage in emerging market firms.
- ▶ In particular, we ask :
 - A. Did the post-crisis build up in leverage affect *firm level distress* in emerging markets?
 - B. If so what are the firm-, industry- and market-wide channels through which corporate leverage might have impacted the credit risk?

In this paper...

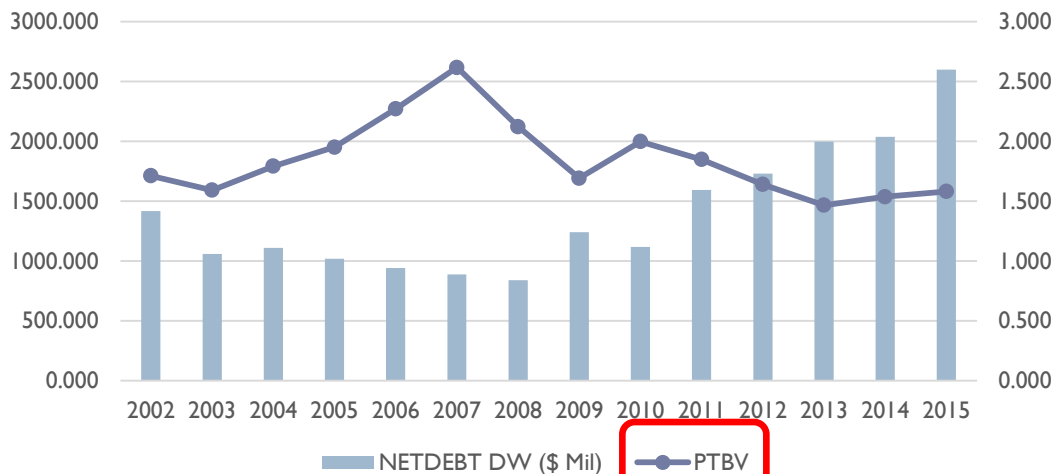
- ▶ We conduct a comprehensive study of how escalating emerging market corporate debt can impact distress risk for issuing firms.
- ▶ Overall our work contributes to the literature on emerging market debt and helps shed light on the ramifications of exploding emerging market debt on underlying credit risk.

Contribution of this paper...

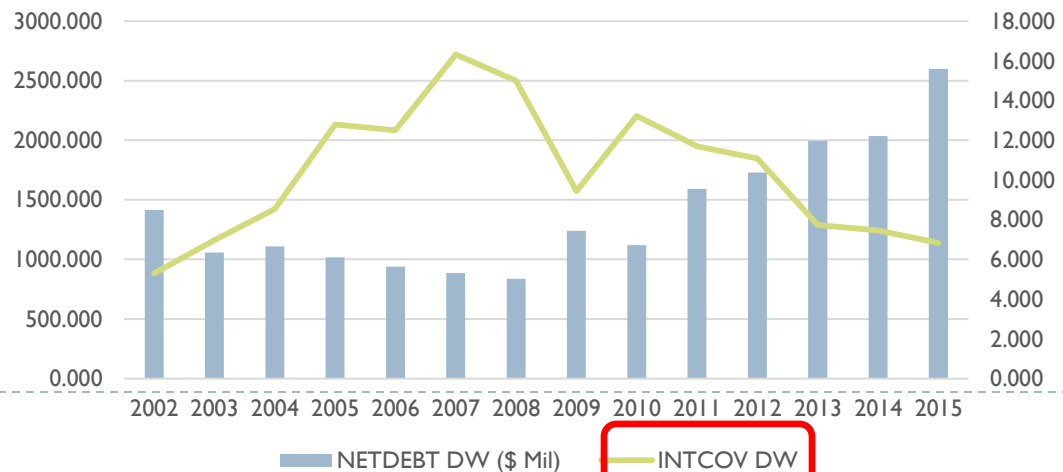
- ▶ Although greater leverage can facilitate higher corporate investment and perhaps stimulate growth,
 - ▶ the continued accumulation of corporate debt can be concerning
 - ▶ because many financial crises in emerging markets have been preceded by rapid leverage growth.
 - ▶ Or potential growth opportunities can mitigate the increased risk premia.
- ▶ To the best of our knowledge this is the first paper
 - ▶ to provide quantification of post-crisis leverage impact on firm level distress and underlying channels of transmission using large scale firm-level emerging market data

Findings 1: Deterioration of firm performance

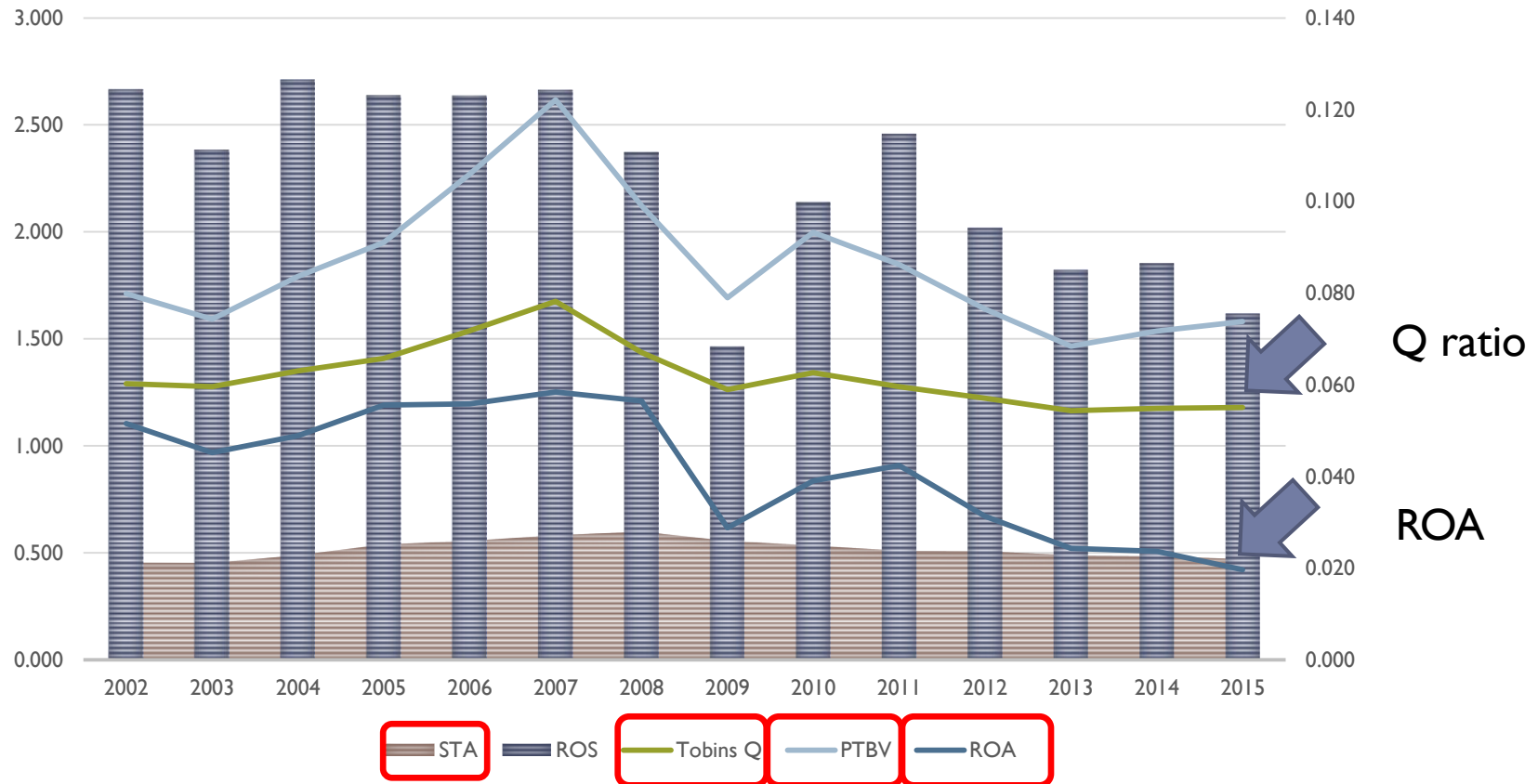
Net debt vs. Market to book



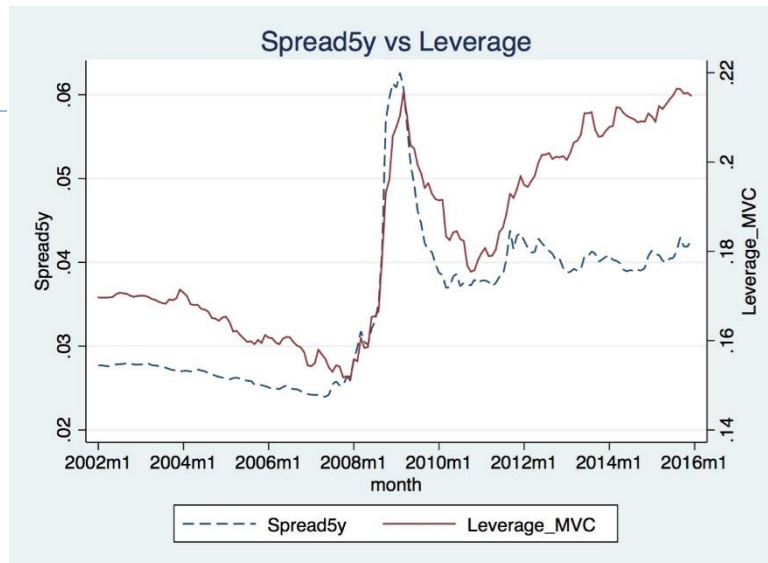
Net debt Vs. Int coverage



Deterioration of firm performance

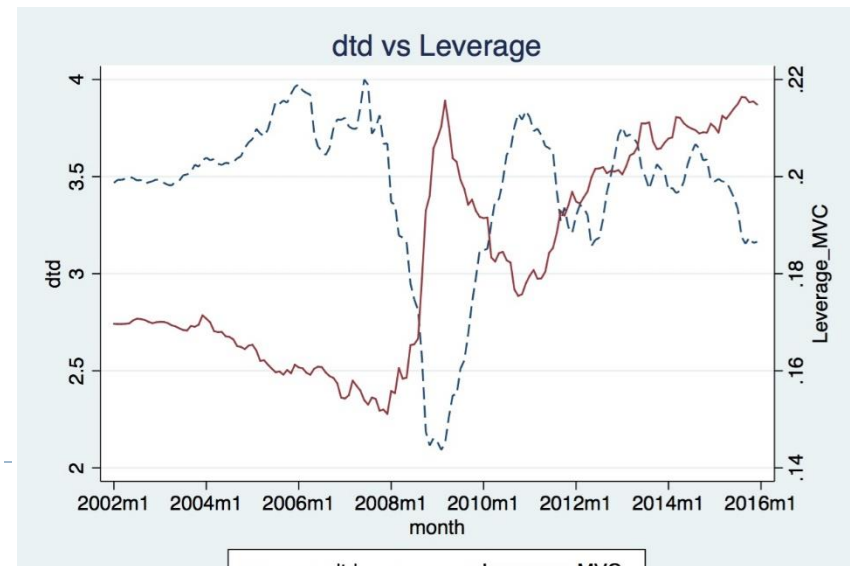
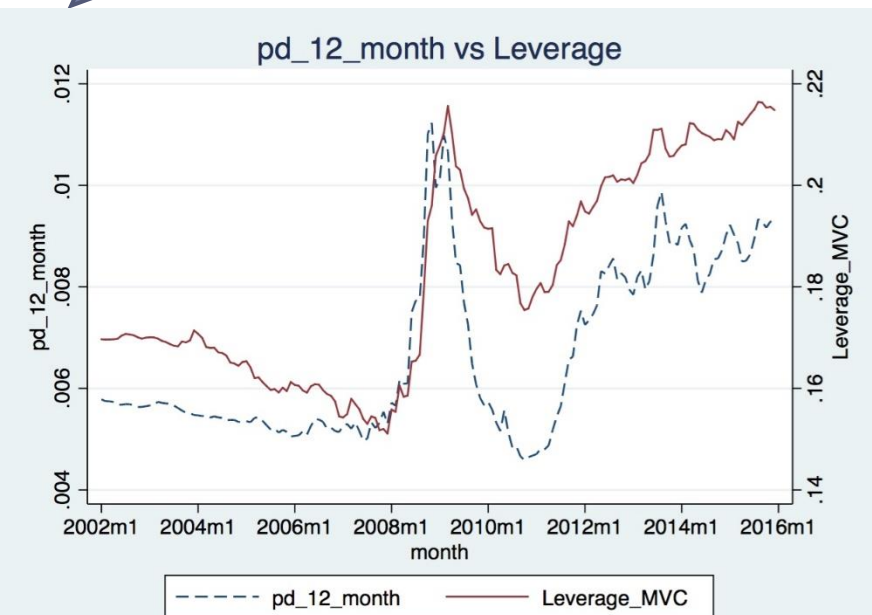
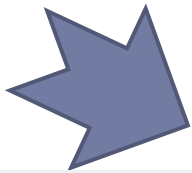


Overall credit risk impact..



While we see no post-crisis rise in CDS spreads...

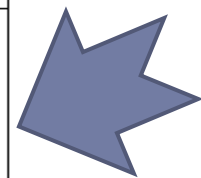
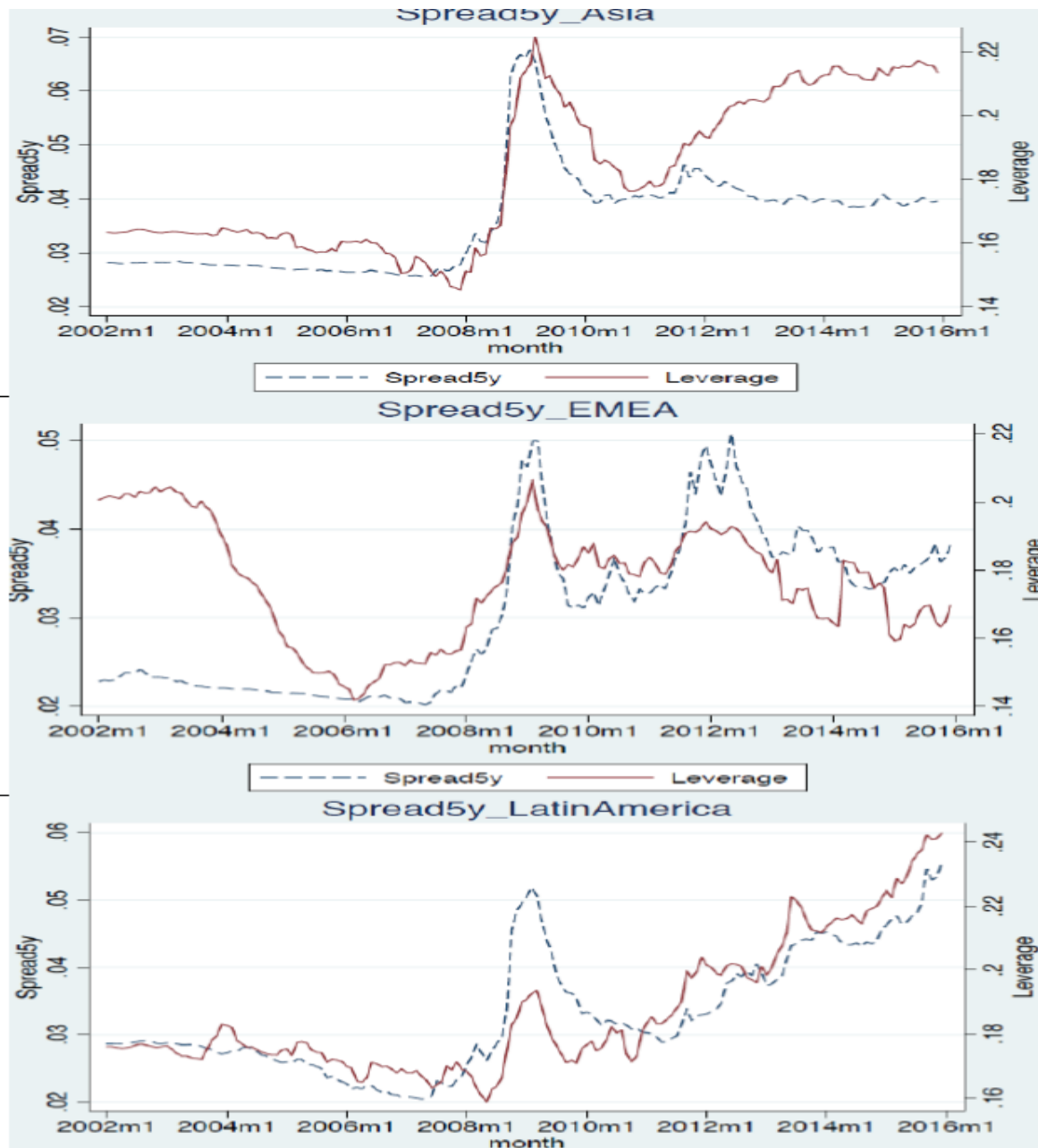
We observe rise in PD and recent drop in DTD post-crisis



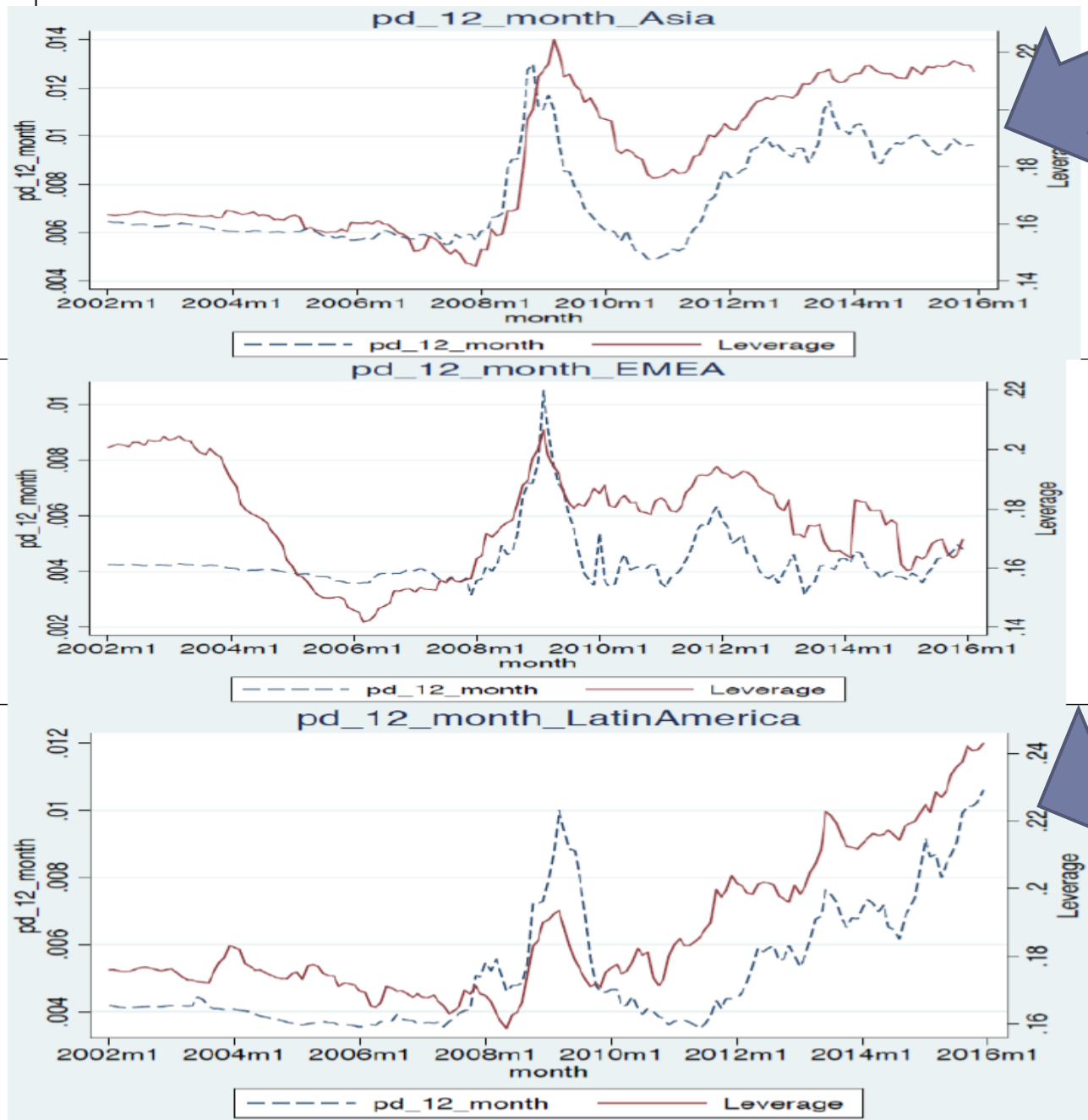
Findings 2: There are strong regional effects

CDS spreads by region

- Cross-sectional differences in leverage
- Steep rise in lev in Asia and Latin America
- We observe rise in CDS only in LA

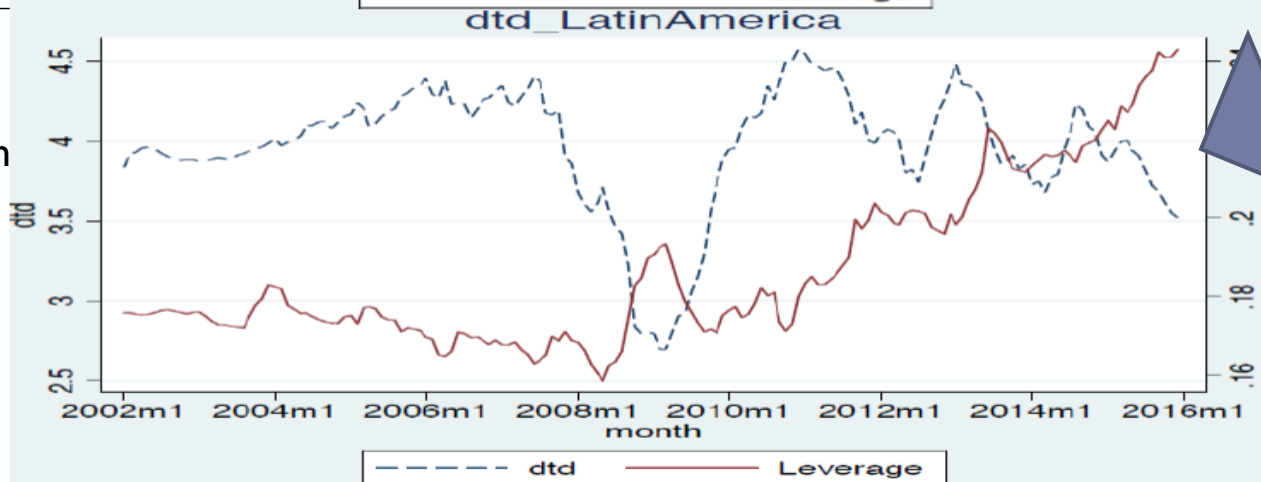
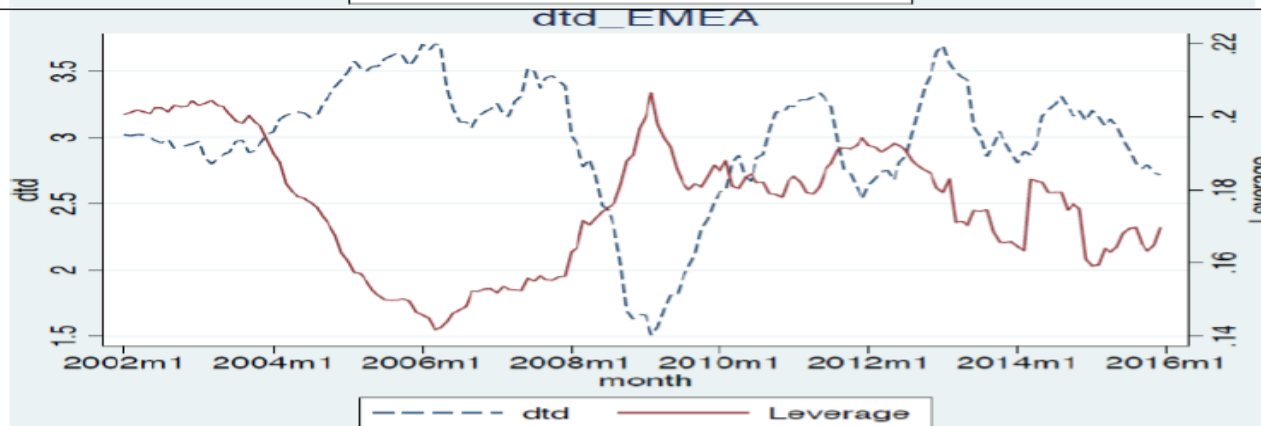
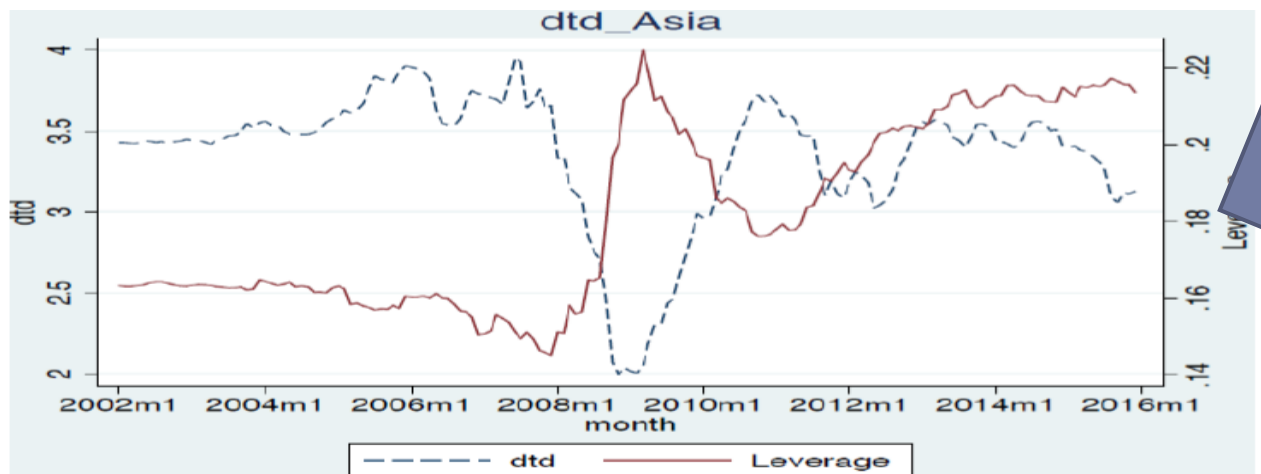


PD by region



We observe rise in PD only in Asia & LA

DTD by region



We observe a drop in DTD only in LA and in Asia recently

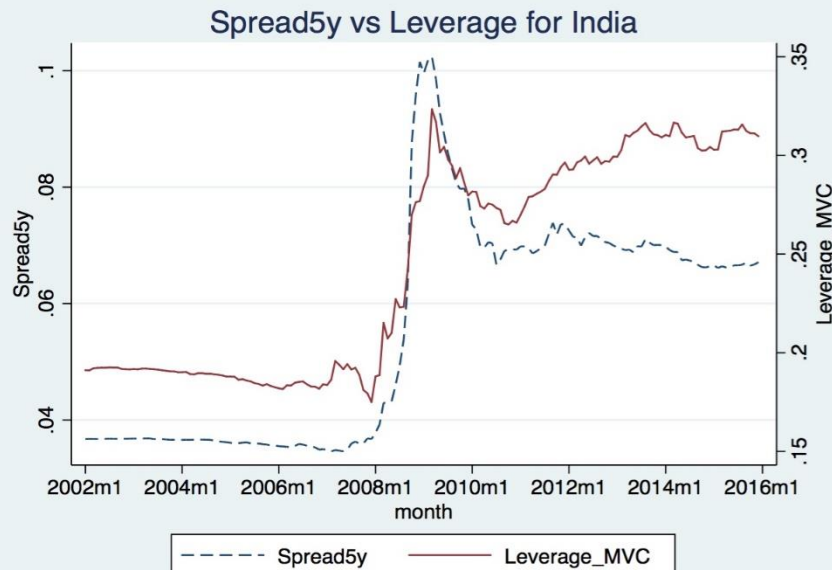
Findings 3: Conditional tests

- ▶ Panel data regressions show that post-crisis increase in leverage has significant impact on
 - ▶ DTD and PD and
 - ▶ the effects are mostly found for Asian (both PD and DTD) and Latin American firms.
- ▶ Leverage also +vely impacts PD slope (proxy for credit risk expectations)
- ▶ Significant importance of aggregate variables such as
 - ▶ local market returns and US term structure slope, VIX and default spreads. (implies a factor structure)
- ▶ CDS spreads become important when we study PCA and averages.
 - ▶ Implies that crosssectionally more leverage does not mean more default.

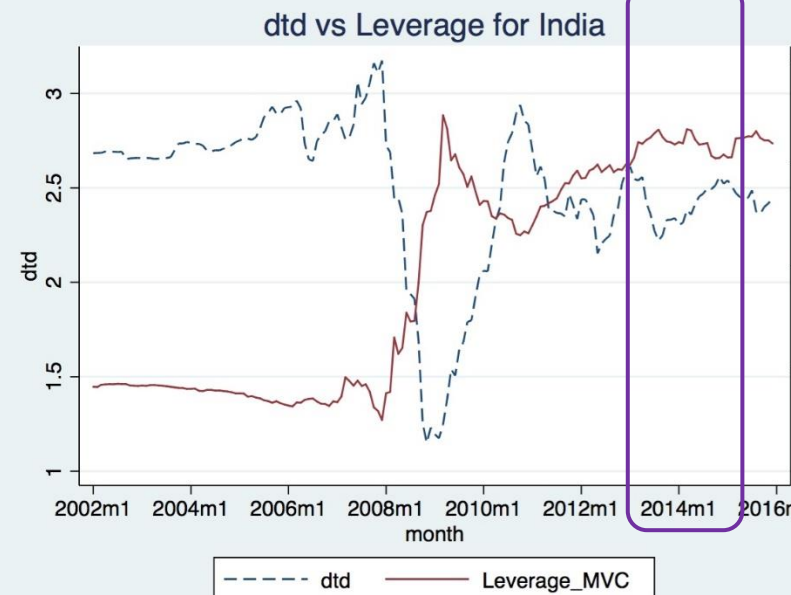
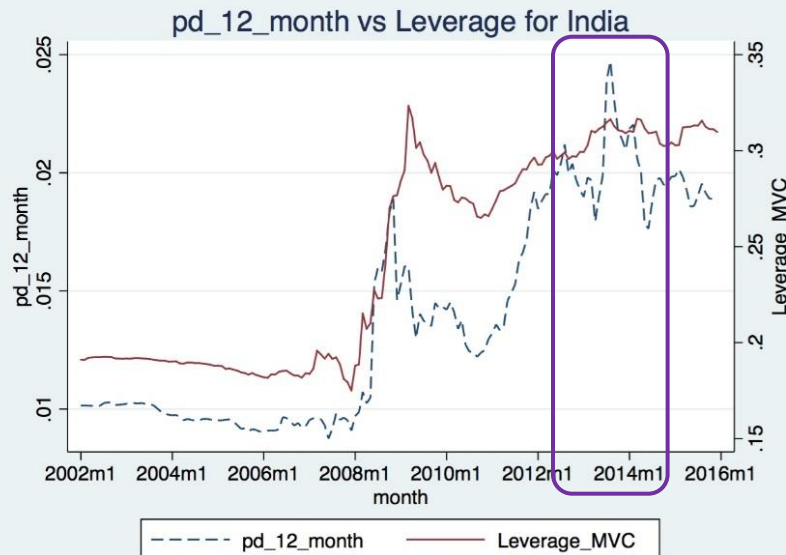
Findings 4: Additionally we find ..

- ▶ Importance of firm and industry specific effects in transmission.
 - ▶ Firm specific variables such as larger size and higher q ratio, signifying high growth prospects, can significantly lower the leverage effect on credit risk.
 - ▶ Leverage impact on credit risk is also significantly higher for tradable industry firms, and remain robust in all three geographic regions.
 - ▶ tradable sector (Industrials, oil and gas, and Technology) non-tradable sectors (construction, transportation, communications, utilities, wholesale/retail trade, and services).
- ▶ Importance of country specific effects in transmission
 - ▶ higher capital flows and favorable exchange rates effect of leverage on credit risk is much lower.
 - ▶ Finally, for firms in countries with better governance effect of leverage on credit risk is significantly lower.

Implications for India



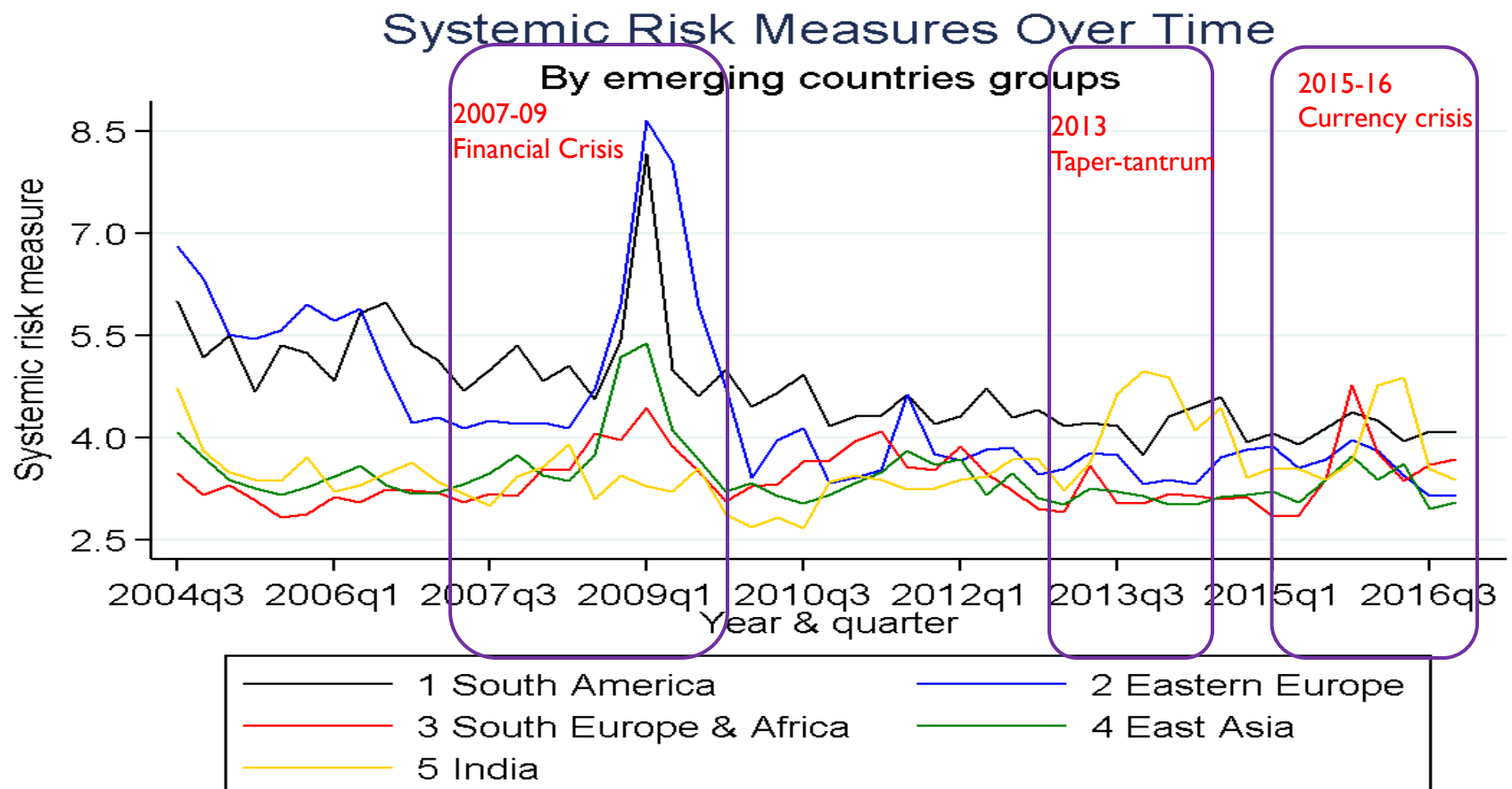
While we see no rise in CDS spreads...



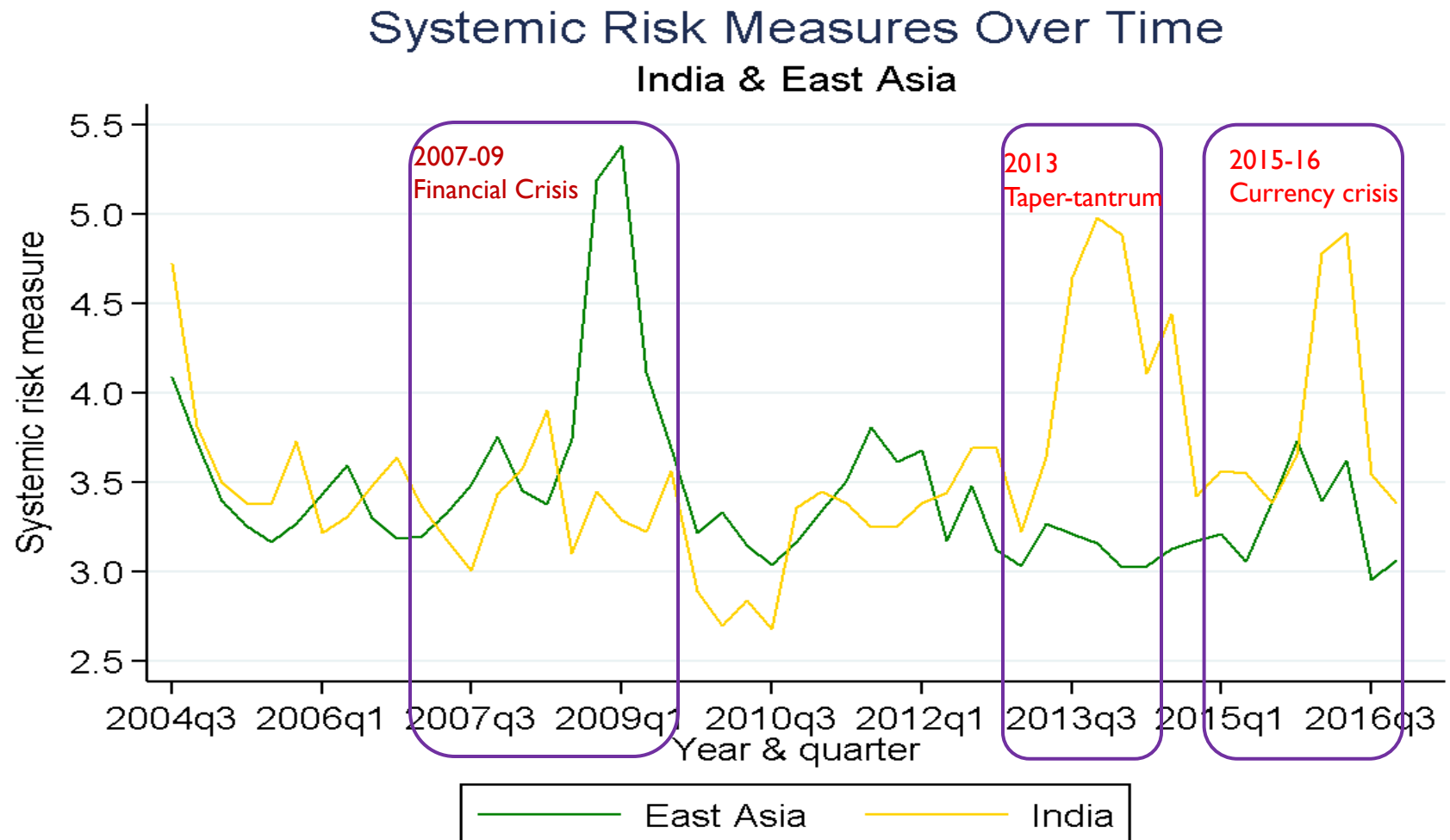
We observe rise in PD post-crisis and more so in PD and DTD during taper-tantrum period in 2013

Our findings have bearings on systemic risk evolution systemic risks in emerging markets

..because systemic risks are a function of default and network risks..



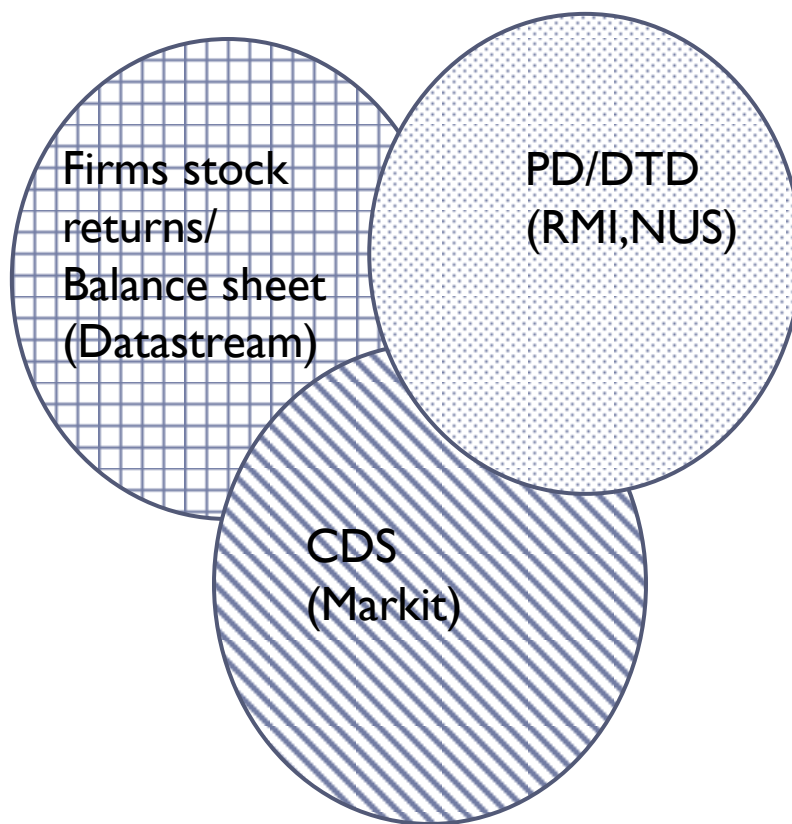
India vs Asia



Key insights

- ▶ Interpretation of credit risk proxies (CDS, PD, DTD)
- ▶ Firm specific covariates are important
- ▶ Using PD as dependent variable

Data sets



From this list, we exclude preference shares and other secondary types of shares issued by companies with the exception is China where we exclude A-shares and include B-shares and H-shares that are accessible to foreign investors.

We identify the list of emerging countries by combining the IMF's & MSCI's lists of emerging countries. Out of the 28 emerging countries from the IMF's and MSCI's lists, 23 emerging countries have CDS data available in Markit database.

Emerging market sample

Home country	Number of companies	% of total number of companies
Argentina	7	2.0%
Brazil	29	8.3%
Bulgaria	1	0.3%
Chile	8	2.3%
China	12	3.4%
Colombia	3	0.9%
Czech Republic	2	0.6%
Egypt	1	0.3%
Greece	7	2.0%
Hungary	3	0.9%
India	74	21.1%
Indonesia	17	4.9%
Malaysia	17	4.9%
Mexico	18	5.1%
Philippines	13	3.7%
Poland	1	0.3%
Russia	22	6.3%
South Africa	11	3.1%
South Korea	34	9.7%
Taiwan	42	12.0%
Thailand	17	4.9%
Turkey	9	2.6%
Ukraine	2	0.6%
TOTAL	350	100.0%

Baseline panel regression (2002-15)

- ▶ Credit risk proxy =
 - ▶ $f(\text{leverage}, \text{leverage} \times \text{crisis}, \text{leverage} \times \text{post-crisis},$
 - ▶ $\text{ldo vol}, \text{ldo vol} \times \text{crisis}, \text{ldo vol} \times \text{post-crisis},$
 - ▶ Controls: balance sheet variables, macro variables)
- ▶ All regressions include controls for year-specific fixed effects, country-specific cluster effects, and t-statistics adjustments for heteroscedasticity, autocorrelation and cross-correlations.

Table VII. Baseline regression

	(1)	(2)	(3)		(4)	(5)	(6)
	Level	Level	Level		Change	Change	Change
VARIABLES	Spread5y	pd_12_month	dtd		Spread5y	pd_12_month	dtd
Leverage_MVC	0.024***	0.011***	-1.545**		0.008*	0.007**	-0.965*
	(0.008)	(0.003)	(0.777)		(0.004)	(0.003)	(0.553)
leverage_crisis	0.010	0.007***	-1.224*		0.004	0.006***	-0.311
	(0.009)	(0.002)	(0.649)		(0.004)	(0.002)	(0.373)
leverage_postcrisis	0.012	0.009**	-1.850**		0.000	0.007***	-0.551
	(0.008)	(0.003)	(0.724)		(0.004)	(0.002)	(0.457)
firmsd_12e	0.141***	0.029***	-11.623***		0.027***	0.010***	-3.780***
	(0.024)	(0.007)	(1.852)		(0.010)	(0.003)	(0.793)
firmsd_12e_crisis	-0.030	-0.006	3.228*		-0.001	-0.004	0.727
	(0.022)	(0.006)	(1.665)		(0.008)	(0.003)	(0.714)
firmsd_12e_postcrisis	-0.031	-0.004	5.688***		-0.011	-0.005	1.209
	(0.025)	(0.009)	(1.991)		(0.009)	(0.005)	(0.811)

Table VII contd.

	(1)	(2)	(3)		(4)	(5)	(6)
	Level	Level	Level		Change	Change	Change
VARIABLES	Spread5y	pd_12_month	dtd		Spread5y	pd_12_month	dtd
INTCOV DW	-0.000	0.000	0.002		-0.000	0.000	0.001
	(0.000)	(0.000)	(0.001)		(0.000)	(0.000)	(0.001)
Sales_to_Assets	-0.001	0.000	0.801***		0.000	0.001	-0.041
	(0.004)	(0.001)	(0.285)		(0.002)	(0.001)	(0.099)
ROS	-0.019	-0.007**	1.255**		-0.000	-0.000	-0.140
	(0.013)	(0.003)	(0.547)		(0.004)	(0.002)	(0.132)
ROA	0.035	0.000	0.778		-0.022*	-0.004	1.090**
	(0.030)	(0.009)	(1.560)		(0.012)	(0.004)	(0.433)
PTBV	-0.003***	-0.002***	0.205***		-0.002***	-0.002***	0.408***
	(0.001)	(0.000)	(0.056)		(0.000)	(0.000)	(0.045)
market_returns	-0.007***	-0.004***	1.085***		-0.006***	-0.001***	0.521***
	(0.002)	(0.001)	(0.115)		(0.001)	(0.000)	(0.046)
sp500	-0.018***	-0.009***	0.708**		-0.004	-0.000	0.091
	(0.006)	(0.002)	(0.340)		(0.003)	(0.001)	(0.117)
vix	0.000***	-0.000***	-0.003		0.000***	0.000***	-0.012***
	(0.000)	(0.000)	(0.004)		(0.000)	(0.000)	(0.002)
default	0.008***	0.002***	-0.769***		0.004***	0.000***	-0.201***
	(0.001)	(0.000)	(0.059)		(0.001)	(0.000)	(0.021)
FRTBS3M	-0.003***	0.000	-0.084*		-0.003***	0.000	0.040*
	(0.000)	(0.000)	(0.047)		(0.000)	(0.000)	(0.021)
slope	-0.002***	-0.000	-0.313***		-0.001*	0.001***	-0.059**
	(0.001)	(0.000)	(0.058)		(0.000)	(0.000)	(0.023)
ted	-0.001	0.001***	-0.217***		0.002***	0.000	0.046***
	(0.001)	(0.000)	(0.050)		(0.000)	(0.000)	(0.015)
Constant	0.011***	0.003**	5.559***		0.000***	0.000*	0.003***
	(0.003)	(0.001)	(0.291)		(0.000)	(0.000)	(0.001)
Observations	16,719	18,261	18,261		16,154	17,921	17,921
Adjusted R-squared	0.387	0.232	0.323		0.139	0.0969	0.131

Robustness tests-

(Leverage X post-crisis) double interaction

ROBUSTNESS CHECKS		Findings
Table VIII	Table VII with Heckman correction	Yes for PD and DTD
Table IX	Table VII for lev and vol quartiles presented selectively for level and changes	Yes for vol groups for PD and DTD No monotonic effects
Table X	PCA analysis use both PD 12 and 60 m	Yes with CDS and DTD Aggregate effect
Table XI	Regression using means use both PD 12 and 60 m	Yes with CDS Aggregate effect
Table XII	Colin Dufresne regressions	Yes PD 10% level No strong effects
Table XIII	Table VII by each region	Asia and Latin America in PD and DTD
Table XIV	Table VII CDS regressions only using modified restructuring CDS data / using alternate credit measures CDS slope and PD slope	

Table VIII

- ▶ We run the first stage probit regressions of
 - ▶ leverage level (high=1; low=0) on several instrumental variables
 - ▶ Sales to Assets ROA, Tobin's Q, all industry dummies, lagged CDS, lagged PD, lagged DTD.
 - ▶ The high (or low) leverage is based on firm's debt being above (or below) on the firm's average leverage ratio for over the sample.
- ▶ We then use inverse Mills ratio (IMR) from the probit model as an additional independent variable in the second stage regression.

Robustness tests:

Table VIII. Endogeneity

VARIABLES	(1) Level Spread5y	(2) Level pd_12_month	(3) Level dtd	(4) Change Spread5y	(5) Change pd_12_month	(6) Change dtd
Leverage_MVC	0.0251*** (0.00845)	0.0118*** (0.00348)	-0.187 (0.670)	0.00563* (0.00320)	0.00504** (0.00245)	-0.671 (0.533)
leverage_crisis	0.00974 (0.00906)	0.00638** (0.00282)	-1.477*** (0.540)	0.00248 (0.00266)	0.00511*** (0.00155)	-0.344 (0.371)
leverage_postcrisis	0.0123 (0.00858)	0.00800** (0.00359)	-2.146*** (0.612)	0.000307 (0.00390)	0.00775*** (0.00257)	-0.664 (0.462)
firmsd_12e	0.142*** (0.0245)	0.0291*** (0.00760)	-11.29*** (1.671)	0.0227** (0.0104)	0.0119*** (0.00332)	-4.311*** (0.937)
firmsd_12e_crisis	-0.0313 (0.0219)	-0.00485 (0.00583)	3.400** (1.552)	0.00192 (0.00697)	-0.00534* (0.00302)	1.044 (0.799)
firmsd_12e_postcrisis	-0.0286	-0.00131	5.028***	-0.00717	-0.00925*	1.613*
IMR	0.000371 (0.00134)	-0.000481 (0.000613)	2.602*** (0.226)	-0.000214* (0.000118)	2.01e-05 (3.24e-05)	-0.0567*** (0.0130)
Constant	0.00963*** (0.00356)	0.00299* (0.00160)	4.216*** (0.278)	0.000362*** (0.000127)	-9.43e-08 (3.62e-05)	0.0614*** (0.0140)
Observations	16,272	16,566	16,566	16,022	16,302	16,302
Adjusted R-squared	0.324	0.132	0.269	0.143	0.0904	0.123

Table X.PCA analysis and averages

Panel A: PCA stacked over with Country and Year fixed effects (first difference)			
	(1)	(2)	(3)
VARIABLES	Spread5y	pd_12_month	dtd
pc1_dcn_Leverage_MVC	-0.0651* (0.0344)	-0.0182 (0.0353)	0.0734 (0.0569)
leverage_crisis	0.130* (0.0751)	0.171** (0.0858)	-0.108* (0.0639)
leverage_postcrisis	0.0996* (0.0594)	0.0624 (0.0480)	-0.144** (0.0662)
pc1_dcn_firmsd_12e	-0.00724 (0.0674)	-0.0334 (0.0394)	-0.0255 (0.0758)
pc1_dcn_firmsd_12e_crisis	0.264** (0.130)	-0.00868 (0.0913)	-0.0894 (0.0846)
pc1_dcn_firmsd_12e_postcrisis	0.0390 (0.0857)	0.0513 (0.0500)	-0.102 (0.0825)
market_return	-0.0686** (0.0289)	-0.0784*** (0.0266)	0.141*** (0.0176)
sp500	-0.104*** (0.0316)	-0.0546* (0.0306)	0.0535*** (0.0200)
dvix	0.110*** (0.0416)	0.0823** (0.0338)	-0.0208 (0.0215)
pc1_dcn_INTCOV_DW	0.000797 (0.0314)	0.0318 (0.0261)	0.0156 (0.0158)
pc1_dcn_PTBV	-0.228*** (0.0418)	-0.303*** (0.0424)	0.370*** (0.0295)
pc1_dcn_ROA	0.0371 (0.0425)	0.0617* (0.0361)	0.00236 (0.0188)
pc1_dcn_ROS	0.0143 (0.0407)	-0.0191 (0.0314)	0.00116 (0.0196)
pc1_dcn_Sales_to_Assets	-0.0144 (0.0309)	0.0317 (0.0366)	0.0147 (0.0165)
arl	0.226*** (0.0526)	0.249*** (0.0427)	0.280*** (0.0263)
Constant	0.00504 (0.156)	-0.0578 (0.137)	-0.0140 (0.105)
Observations	2,559	2,559	2,559
Adjusted R-squared	0.401	0.425	0.558

Panel B: Row Means Stacked with country and year fixed effects (1st Difference)			
	(1)	(2)	(3)
VARIABLES	Spread5y	pd_12_month	dtd
Avg_Leverage_MVC	-0.016*** (0.005)	-0.001 (0.002)	-1.815 (1.361)
leverage_crisis	0.055*** (0.021)	0.017*** (0.005)	2.072 (1.677)
leverage_postcrisis	0.037*** (0.014)	0.005 (0.004)	-0.180 (1.568)
Avg_firmsd_12e	-0.012 (0.010)	0.008 (0.007)	0.479 (4.293)
Avg_firmsd_12e_crisis	0.136* (0.078)	0.005 (0.021)	-4.323 (4.735)
Avg_firmsd_12e_postcrisis	0.030 (0.034)	-0.005 (0.010)	-6.135 (4.490)
market_return	-0.004*** (0.001)	-0.002*** (0.000)	0.722*** (0.101)
sp500	-0.013*** (0.004)	-0.003*** (0.001)	0.294 (0.240)
dvix	0.000 (0.000)	-0.000 (0.000)	-0.001 (0.002)
Avg_INTCOV_DW	0.000 (0.000)	0.000** (0.000)	-0.000 (0.003)
Avg_PTBV	-0.005*** (0.001)	-0.002*** (0.000)	0.615*** (0.082)
Avg_ROA	-0.039 (0.027)	-0.004 (0.008)	1.785 (1.457)
Avg_ROS	0.004 (0.008)	0.004 (0.003)	0.144 (0.416)
Avg_Sales_to_Assets	-0.003 (0.008)	0.001 (0.002)	0.508 (0.339)
arl	0.195*** (0.069)	0.252** (0.103)	0.231*** (0.053)
Constant	0.000 (0.000)	-0.000 (0.000)	0.004 (0.016)
Observations	2,559	2,559	2,559
Adjusted R-squared	0.270	0.257	0.312

Table XI. Cross-sectional averages of monthly time-series regressions

VARIABLES	Δ CDS		Δ PD		Δ DTD	
	(1)	(2)	(3)	(4)	(5)	(6)
Leverage_MVC	0.014*** (0.005)	0.019* (0.011)	0.017*** (0.003)	0.010 (0.007)	-2.605*** (0.477)	-1.721 (1.295)
leverage_X_crisis		0.014 (0.013)		0.012* (0.006)		-3.082 (1.956)
leverage_X_postcrisis		-0.005 (0.013)		0.012* (0.007)		-1.061 (1.158)
firmsd_12e	0.005 (0.007)	0.017 (0.012)	0.001 (0.003)	0.001 (0.005)	-4.801*** (0.944)	-4.214*** (1.646)
firmsd_12e_X_crisis		-0.032* (0.017)		0.005 (0.007)		-6.329 (5.495)
firmsd_12e_X_postcrisis		-0.002 (0.017)		-0.004 (0.006)		-0.780 (2.028)
INTCOV_DW	-0.001 (0.001)	-0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.015 (0.014)	0.002 (0.015)
PTBV	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	1.012*** (0.079)	1.042*** (0.083)
ROA	0.132 (0.353)	0.255 (0.388)	0.255 (0.415)	0.292 (0.412)	-29.719 (49.485)	-45.206 (49.684)
ROS	-0.029 (0.046)	-0.064 (0.054)	-0.037 (0.042)	-0.041 (0.041)	6.805 (5.484)	7.199 (5.549)
Sales_to_Assets	-0.029 (0.073)	-0.037 (0.078)	-0.075 (0.079)	-0.083 (0.079)	4.905 (11.402)	8.417 (11.511)
market return	-0.006*** (0.001)	-0.006*** (0.001)	-0.001 (0.001)	-0.001 (0.001)	0.306*** (0.072)	0.305*** (0.075)
sp500	0.003 (0.003)	0.004 (0.003)	-0.001 (0.001)	0.001 (0.001)	-0.081 (0.148)	-0.045 (0.154)
vix	0.001*** (0.001)	0.001*** (0.001)	0.001* (0.001)	0.001* (0.001)	-0.010*** (0.002)	-0.010*** (0.002)
def	0.001 (0.001)	0.001 (0.001)	0.001** (0.001)	0.001* (0.001)	-0.105*** (0.032)	-0.113*** (0.032)
3mTBill	-0.001 (0.001)	-0.001 (0.001)	0.002*** (0.001)	0.001*** (0.001)	-0.016 (0.101)	-0.013 (0.101)
term	-0.001 (0.001)	-0.001 (0.001)	0.001*** (0.001)	0.001*** (0.001)	-0.079*** (0.030)	-0.069*** (0.031)
TED	0.005*** (0.001)	0.004*** (0.001)	0.001 (0.001)	0.001 (0.001)	0.011 (0.084)	0.013 (0.086)
Constant	0.001 (0.001)	0.001* (0.001)	0.001 (0.001)	0.001 (0.001)	0.005* (0.003)	0.004 (0.003)
Observations	97.31	97.31	97.99	97.99	97.99	97.99
Adjusted R-squared	0.474	0.496	0.449	0.481	0.384	0.419

Table XIII. Panel regressions of credit risk by region

VARIABLES	Asia			EMEA			Latin America		
	ΔCDS	ΔPD	ΔDTD	ΔCDS	ΔPD	ΔDTD	ΔCDS	ΔPD	ΔDTD
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Leverage_MVC	0.009* (0.005)	0.007* (0.003)	-0.529 (0.550)	-0.003 (0.009)	-0.001 (0.008)	-4.402*** (0.699)	0.001 (0.012)	0.005 (0.003)	-1.079 (1.005)
leverage X crisis	0.005 (0.004)	0.006*** (0.002)	-0.471 (0.348)	0.001 (0.005)	-0.005* (0.003)	2.006*** (0.548)	-0.000 (0.019)	0.015*** (0.004)	-1.598 (1.224)
leverage X postcrisis	-0.002 (0.005)	0.007** (0.003)	-0.983** (0.430)	0.014 (0.013)	0.008 (0.010)	3.029*** (0.599)	0.001 (0.009)	0.013*** (0.005)	-0.742 (1.210)
firmsd_12e	0.030** (0.013)	0.008** (0.004)	-3.834*** (0.867)	0.004 (0.016)	-0.006 (0.008)	0.817 (1.524)	0.026 (0.017)	0.026*** (0.008)	-5.422** (2.398)
firmsd_12e X crisis	-0.001 (0.008)	-0.005 (0.004)	0.781 (0.735)	-0.040** (0.017)	0.015* (0.008)	-3.802** (1.522)	0.013 (0.028)	-0.013 (0.008)	3.285 (2.338)
firmsd_12e X postcrisis	-0.013 (0.010)	-0.010* (0.006)	1.754** (0.838)	-0.043 (0.029)	0.015 (0.014)	-5.061*** (1.636)	0.013 (0.019)	-0.006 (0.010)	1.854 (2.480)
Constant	0.000*** (0.000)	0.000 (0.000)	0.005*** (0.001)	0.000 (0.000)	-0.000 (0.000)	-0.005*** (0.001)	0.000*** (0.000)	0.000*** (0.000)	0.002 (0.002)
Observations	11,054	12,174	12,174	1,930	2,260	2,260	3,170	3,487	3,487
Adjusted R-squared	0.161	0.114	0.134	0.0688	0.0813	0.198	0.148	0.108	0.122

Channels of transmission?

Firm/Industry channels

CHANNELS OF TRANSMISSION		FINDINGS
Table XIV.A	Effect of short term debt on credit risk Rerun Table VII using short term debt - Short-term Leverage X post-crisis	- Lowers DTD - No incremental effects for short term debt.
Table XIV. B	Effect of firm size on credit risk - Leverage X post-crisis X high/low size dummy	- Larger firms from top quartile have lower PD compared to bottom three quartiles
Table XIV. C	Effect of growth factor (Q) on credit risk - Leverage X post -crisis X high/low Q dummy	- Larger Q ratio firms from top quartile have lower CDS and PD compared to bottom three quartiles. - Larger Q ratio firms from top quartile lower PD compared to bottom quartile
Table XIV. D & E	Effect of Industry effects and (industry X region) effects - Leverage X post-crisis X tradeables dummy tradeable sector dummy i.e. 1 for tradeable sector (Industrials, oil and gas, and Technology) and zero for non-tradeable sectors (construction, transportation, communications, utilities, wholesale/retail trade, and services).	- Tradeable firms higher PD compared to non-tradeables - Tradeable firms higher PD and lower DTD compared to non-tradeables for all the three regions

Market wide channels

	CHANNELS OF TRANSMISSION	FINDINGS
Table XV.A	Effects of capital flows and external debt - Leverage X post-crisis Capflows high/low dummy Leverage X post-crisis external debt high/low dummy	- Low Capflows increases credit risk (CDS and PD) compared to bottom three quartiles. - <i>External debt no conclusive results</i>
Table XV. B	-Effect of FOREX (USD/local) -Leverage X post-crisis x FOREX high/low dummy	- Standalone strong local rates lower credit risk - Triple interaction shows strong forex lower credit risk and vice versa (PD and DTD)
Table XV. C	Effects of country level governance on credit risk -Leverage X post-crisis x GOV factor high/low dummy	- Strong governance lowers PD

Takeaway

- ▶ In this paper, we assess the potential fallout from the increased corporate leverage post-crisis in emerging market firms.
- ▶ We study how escalating emerging market corporate debt can impact the credit risk of underlying firms (first paper to do so!)
- ▶ Our panel regression results show that in the post crisis period, higher leverage significantly increases the PD of the firms; and lowers the DTD.
- ▶ The incremental leverage effects are most prominent in the Asian and Latin American region firms.
- ▶ We also identify several channels of transmission of the leverage impact.



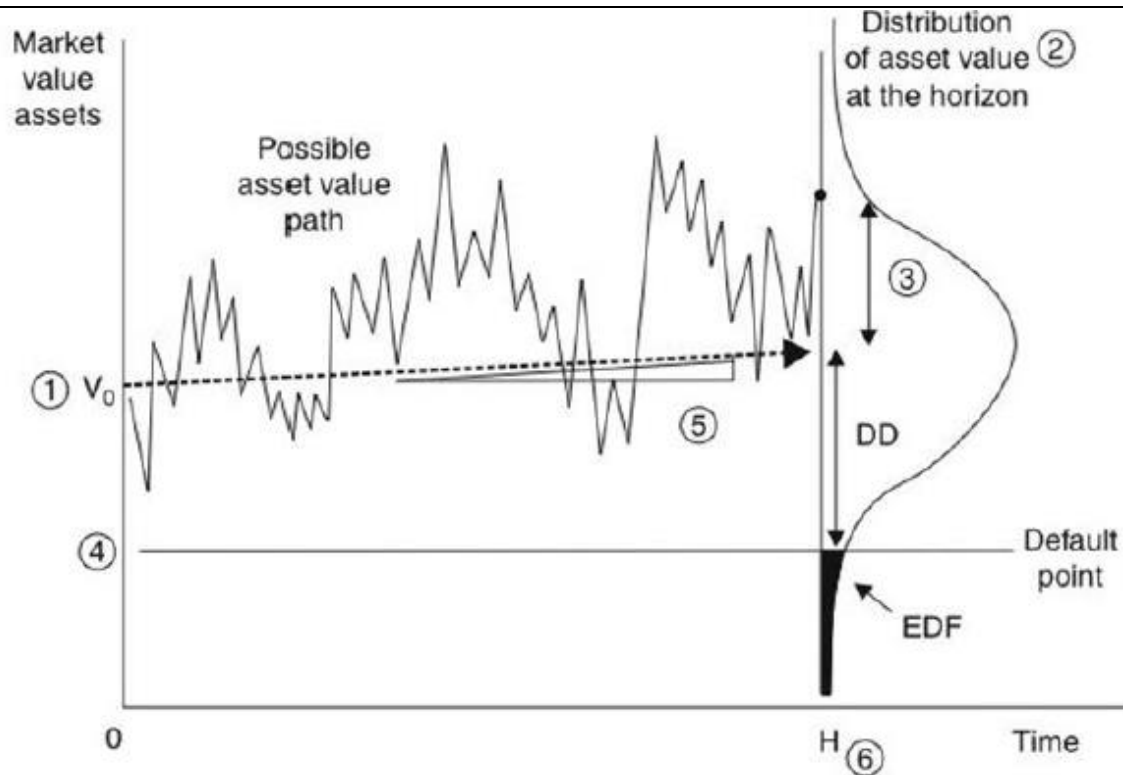
Appendix



$$DD = \frac{\text{value of firm's assets} - \text{face value of debt}}{\text{volatility of firm value}}$$

- The firm value and firm value volatility are unobserved variables and need to be solved for.
- See the next slide for a possible iterative procedure.

PD vs DD



CDS Vs PD

$$S = \ln(1 - P) \frac{R - 1}{t}$$

- S is the spread expressed in percentage terms (not basis points)
- t are the years to maturity
- R is the recovery rate in percentage terms

Merton (1974) Model

credit spreads. The value of risky debt in Equation 10.4 can be written as follows to emphasize the role of probability of default and loss given default:

$$\begin{aligned}
 \text{Risky bond } D(V_t, t) &= Fe^{-r(T-t)} - Fe^{-r(T-t)} + Fe^{-r(T-t)} N(d_2) + V_t N(-d_1) \\
 &= Fe^{-r(T-t)} - Fe^{-r(T-t)} N(-d_2) + V_t \frac{N(-d_2)}{N(-d_2)} N(-d_1) \\
 &= Fe^{-r(T-t)} - \left[Fe^{-r(T-t)} - V_t \frac{N(-d_1)}{N(-d_2)} \right] N(-d_2). \quad (10.6) \\
 &= Fe^{-r(T-t)} (1 - N(-d_2)) + V_t N(-d_1) \\
 &= V_t N(-d_1) + Fe^{-r(T-t)} (1 - N(-d_2)) \\
 &= V_t N(-d_1) + Fe^{-r(T-t)} (N(d_2))
 \end{aligned}$$

PV of expected Loss
CDS value

Prob of default

Prob of no default

PD: Forward intensity model by Credit Risk Initiative

- ▶ Source: Credit Risk Initiative (CRI), NUS
- ▶ The underlying forward intensity model was introduced in Duan, Sun and Wang (2012) and also borrows from Duan and Fulop (2013).
- ▶ This model allows probability of default (PD) forecasts to be made at a range of horizons.
 - ▶ The forward intensity model is a reduced form model in which the PD is computed as a function of different input variables. These can be firm-specific or common to all firms within an economy.
- ▶ The model improves on the earlier reduced form model by Duffie, Saita and Wang (2007) that relies on modeling the time series dynamics of the input variables in order to make PD forecasts for different horizons.
- ▶ Given the complexity in modeling the dynamics of variables such as accounting ratios,
 - ▶ Duffie et al., (2007) model is difficult to implement if different forecast horizons are required.

Covariates (Duan, Sun and Wang, 2012)

We use the following set of common factors and firm-specific attributes to characterize the forward intensity functions.

1. SP500: trailing 1-year return on the S&P500 index.
2. Treasury rate: 3-month US Treasury bill rate.
3. DTD: firm's distance-to-default, which is a volatility adjusted leverage measure based on Merton (1974). The DTD is estimated once a month using the preceding one year of daily equity values. In order to include financial firms in our analysis, we follow Duan's (2010) adjustment method to add a fraction of firm's other liabilities to the default point, which is by the well-known KMV formula defined as the sum of 100% short-term debt and 50% long-term debt. The model's parameters are estimated by the transformed-data maximum likelihood method in Duan (1994, 2000), and plugged into the DTD formula. We take the last DTD in the one-year series. The methodological details are provided in Appendix B.
4. CASH/TA: the ratio of the sum of cash and short-term investments to the total assets.
5. NI/TA: the ratio of net income to the total assets.
6. SIZE: logarithm of the ratio of a firm's market equity value to the average market equity value of the S&P500 firms.
7. M/B: the market-to-book asset ratio.
8. SIGMA: 1-year idiosyncratic volatility, calculated by regressing individual monthly stock return on the value-weighted CRSP monthly return over the preceding 12 months. SIGMA is the standard deviation of the residuals from the regression. Following Shumway (2001), we treat SIGMA as missing if there

Key variables...

Panel A: Credit risk variables

CDS spreads	5-year CDS spreads at the end of the month (EOM) and monthly mean spreads (mean) calculated as the mean of daily spreads. Daily 5-year CDS spread are obtained from Markit database.
CDS liquidity	The number of unique contributors (Composite Depth) for the 5-year CDS spreads at the end of the month (EOM) and monthly mean values. Daily data are obtained from Markit database.
CDS volatility	Estimated using two proxies: (a) the range (maximum minus minimum) of 5-year CDS spreads over the month, and (b) standard deviation of historical 6 month daily 5-year CDS spread are obtained from Markit database
CDS slope	The difference between 10-year and 2-year CDS spreads at the end of the month (EOM) and monthly mean values. Daily 10-year and 2-year CDS spread are obtained from Markit database.
PD	12-month and 60-month probability of default, obtained from the Credit Risk Initiative (CRI) at the Risk Management Institute (RMI) of the National University of Singapore (NUS).
PD slope	Difference between 60 month and 12 month CDS spread.
DTD	Monthly distance-to-default data, obtained from the Credit Risk Initiative (CRI) at the Risk Management Institute (RMI) of the National University of Singapore (NUS).
Equity market risk	Three metrics based on market adjusted individual firm returns: Std dev, skewness, and kurtosis.

Leverage variables

Panel B: Firm-level variables

Market value of equity	Market value of company at the end of the month, obtained from Datastream.
Net debt DW	Net debt represents Total debt minus Cash & Short-term investments, 12-month trailing values are calculated by Datastream based on Worldscope (DW) data; obtained from Datastream.
Leverage	The ratio of Net Debt to Total Assets. Total assets 12-month trailing values are calculated by Datastream based on Worldscope data; obtained from Datastream.
Leverage_MVC	The ratio of Net Debt to Market value of assets: $[\text{Net Debt} / (\text{Total Assets} - \text{Book value of equity} + \text{Market value of equity})]$. Total Assets and Book Value of Equity are 12-month trailing values that are calculated by Datastream based on Worldscope data; obtained from Datastream.
Debt/EBIT	The ratio of Net debt to EBIT. EBIT 12-month trailing values are calculated by Datastream based on Worldscope data; obtained from Datastream.
Debt/market value of equity	The ratio of Net debt to Market value of company.
Debt/book value of equity	The ratio of Net debt to Book value of equity. Book value of equity 12-month trailing values are calculated by Datastream based on Worldscope data; obtained from Datastream.

Panel B: Firm-level variables

PTBV	Price-to-book value ratio, obtained from Datastream.
Interest coverage DW	Interest charge coverage is the ratio of EBIT to Total Interest Expense, 12-month trailing values are calculated by Datastream based on Worldscope (DW) data; obtained from Datastream.
STA	The ratio of company's Net Sales to Total Assets. Net sales 12-month trailing values are calculated by Datastream based on Worldscope data; obtained from Datastream. Total assets 12-month trailing values are calculated by Datastream based on Worldscope data; obtained from Datastream.
ROS	The ratio of company's Net Income to Net Sales. Net income 12-month trailing values are calculated by Datastream based on Worldscope data; obtained from Datastream. Net sales 12-month trailing values are calculated by Datastream based on Worldscope data; obtained from Datastream.
ROA	The ratio of company's Net Income to Total Assets. Net income 12-month trailing values are calculated by Datastream based on Worldscope data; obtained from Datastream. Total assets 12-month trailing values are calculated by Datastream based on Worldscope data; obtained from Datastream.
Tobin's Q	The ratio of market value of assets to book value of assets: $[(\text{Total Assets} - \text{Book value of equity} + \text{Market value of equity}) / \text{Total Assets}]$. Total Assets and Book Value of Equity are 12-month trailing values that are calculated by Datastream based on Worldscope data; obtained from Datastream.

Key variables contd.

Panel D: Country-level variables

S&P 500	Marker returns using the S&P 500 index
def	Default factor, obtained as Moody's BAA yield minus 10-year swap rate.
level	Level factor obtained as 3 month T-Bill rate
term	Term structure factor, obtained as 10-year rate minus 2-year Treasury rates.
vix	Equity market volatility factor, obtained as VIX index.
TED	Aggregate liquidity factor, obtained as 30-day LIBOR rate minus 3-month Treasury-Bill rate.
US-Swap rate	Swap rate for U.S. dollar
country_ret vol	Country-specific equity return volatility, computed as standard deviation of country-specific monthly stock index returns over the year preceding the bond issuance date.
non-fdi	Capital flows is captured using non-foreign direct investment net capital which measures the monetary value of capital inflow net of capital outflow other than foreign direct investment. (source: Oxford Economics, Datastream)
netdebt	External debt is measured as the outstanding amount of debt owed to non-residents as a % of GDP. (source: Oxford Economics, Datastream)

Countries in each Region

Panel F: Emerging market economies

Asia	Includes China, India, Indonesia, Korea, Malaysia, Phillipines, Taiwan, Thailand, and Russian Federation.
EMEA	Refers to Eastern Europe, Middle East and Africa, and includes Bulgaria, Czech Republic, Egypt, Greece, Hungary, Turkey, Poland, South Africa, and Ukraine.
Latin America	Includes Argentina, Brazil, Chile, Colombia, and Mexico.